

## How I use the HPC cluster to run my code

- Connecting
  - `ssh youronid@submit.hpc.engr.oregonstate.edu`
  - VERY similar to connecting to the flip server
- Running jobs
  - I mostly submitted batch jobs - this was nice because it emailed me when the job started and ended (as well as if it failed or completed successfully). So, I didn't have to wait around for it to complete!
  - This page has most of the info for submitting the jobs, there are other ways to submit jobs included as well.  
<https://it.engineering.oregonstate.edu/hpc/slurm-howto>
  - An example of the bash script to submit your job:

```
#!/bin/bash
#SBATCH -J helloWorld          # name of job
#SBATCH -A mySponsoredAccount  # name of my sponsored
account, e.g. class or research group, NOT ONID!
#SBATCH -p share               # name of partition or queue
#SBATCH -o helloWorld.out      # name of output file for
this submission script
#SBATCH -e helloWorld.err      # name of error file for
this submission script
#SBATCH --mail-type=BEGIN,END,FAIL
#SBATCH --mail-user=joeparallel@oregonstate.edu

# load any software environment module required for app (e.g.
matlab, gcc, cuda)
module load software/version

# run my job (e.g. matlab, python)
mySoftwareExecutable
```
  - Note that you may need a specific sponsored account or partition access, this is where I had to reach out to OSU IT (Robert Yelle) to get added to the correct ones. I believe the “share” partition is available to anyone (there is a list of the partitions available in the link above).
- Helpful Resources
  - <https://it.engineering.oregonstate.edu/hpc>
  - Robert Yelle is the main contact for anything HPC related. If you fill out a ticket through OSU IT about the HPC it will be routed to him.
  - Mike Bailey's CS 475 slides:  
[https://web.engr.oregonstate.edu/~mjb/cs575/Handouts/dgx\\_system.1pp.pdf](https://web.engr.oregonstate.edu/~mjb/cs575/Handouts/dgx_system.1pp.pdf)  
These instructions are mainly focused on running c++ scripts, but helped me understand the HPC more.